Fall 2019 VE496
Lecture notes
Week 6

- Outlines
- Headings
- COPE skills for technical & professional writing

Benefits of Outlining

A good outline shows:

- what things go together **(grouping)**
- what order they should be in **(ordering)**
- how the ideas relate **(hierarchy)**

For longer reports, your outline will be the basis of your **table of contents**

Conventional outlining system

Roman numerals
Alphabet letters
(Upper & lower cases)
Arabic numbers

I. First-level heading
   A. Second level, first part
   B. Second level, second part
      1. Third level, first part
      2. Third level, second part
         a. Fourth level, first part
            (1) Fifth level, first part
               (a) Sixth level, first part
II. First-level heading
   A. Second level, first part
   B. Second level, second part
   etc.

Decimal outlining system

Whole numbers
Decimals

1.0 First-level heading
- 1.1 Second level, first part
- 1.2 Second level, second part
  - 1.2.1 Third level, first part
  - 1.2.2 Third level, second part
    - 1.2.2.1 Fourth level, first part
      - 1.2.2.1.1 Fifth level, first part
        - 1.2.2.1.1.1 Sixth level, first part

2.0 First-level heading
- 2.1 Second level, first part
- 2.2 Second level, second part
- etc.

Two kinds of Headings

Descriptive (Topical) headings (1-2 words) *Only identify the topic*	**Informative (Talking) headings** (longer) *Also summarize key point*
Cost	Increase in Cost of Operation
Space Requirement	Less Space Required

DESCRIPTIVE (TOPICAL) OUTLINE

1. Industry Characteristics
    a. Annual sales
    b. Profitability
    c. Growth rate
        i. Sales
        ii. Profit

Informative (talking) outline

Summary Form

1. Flour milling is a mature industry.
    a. Market is large.
    b. Profit margins are narrow.
    c. Growth is modest.
        i. Sales growth averages less than 3 percent a year.
        ii. Profits are flat.

DESCRIPTIVE (TOPICAL) OUTLINE

1. Industry Characteristics
    a. Annual sales
    b. Profitability
    c. Growth rate
        i. Sales
        ii. Profit

✓ Better for readers who want to see the facts before being told what to think about them.

Informative (talking) outline

Summary Form

1. Flour milling is a mature industry.
   a. Market is large.
   b. Profit margins are narrow.
   c. Growth is modest.
      i. Sales growth averages less than 3 percent a year.
      ii. Profits are flat.

✓ Better if your readers are busy, trust your judgment, and are likely to skim the supporting facts

Topical heading & talking heading

1. Parallel construction
2. Precise wording
3. Variety in wording

1. Parallel construction in headings

The 3 headings below illustrate violations of parallelism:

1. Programmer Output Is Lagging (sentence).
2. Increase in Cost of Labor (noun phrase)
3. Unable to Deliver Necessary Results (adjective phrase)

Corrections:

1. Lag in Programmer Output
2. Increase in Cost of Labor
3. Inability to Deliver Necessary Results

1. Programmer Output Is Lagging.
2. Cost of Labor Is Increasing.
3. Information Systems Cannot Deliver Necessary Results.

Exercise:

1. Managers Prefer an Intranet
2. U.S. Employees Prefer a Social Media Site
3. A Newsletter Is Preferred by Overseas Employees

Correction:

Overseas Employees prefer a Newsletter

1. Parallel construction in headings
2. **Precise wording in headings**

A. Personal Appearance Enhancement Is the Most Desirable Feature of Contact Lenses That Wearers Report.

B. The Drawback of Contacts Mentioned by Most People Who Can’t Wear Them Is That They Are Difficult to Put in.

C. More Comfort Is the Most Desired Improvement Suggested by Wearers and Non-wearers of Contact Lenses.

A. Personal Appearance Enhancement Is the Most Desirable Feature of Contact Lenses That Wearers Report.

A. Most Desirable Feature: Personal Appearance

B. The Drawback of Contacts Mentioned by Most People Who Can’t Wear Them Is That They Are Difficult to Put in.

B. Prime Criticism: Difficulty of Insertion

C. More Comfort Is the Most Desired Improvement Suggested by Wearers and Non-wearers of Contact Lenses.

C. Most Desired Improvement: Comfort

1. **Parallel construction heading**
2. **Precise wording in heading**
3. **Variety in wording in heading**

Oil Production in Texas

Oil Production in California

Oil Production in Louisiana

Texas Leads in Oil Production

California Holds the Runner-up Position

Rapidly Gaining Louisiana Ranks Third

Using headings in your table of contents

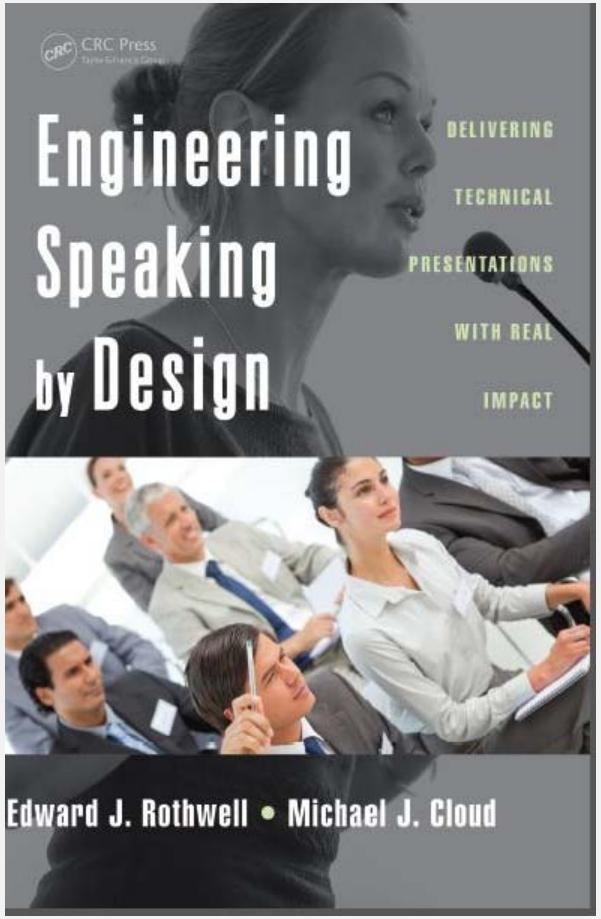
The table of contents indicates in outline form:

- Coverage
- Sequence
- Relative importance

of the information in your report

- Simplicity vs. completeness
- Length & complexity of report

Week 7 & 8 reading



Stereotypes about engineers:

- You have no social skills.
- The typical dork/geek stereotypes (studious but boring)
- The English language is your enemy… you cannot spell, let alone write very coherently.
- You have no fashion sense.

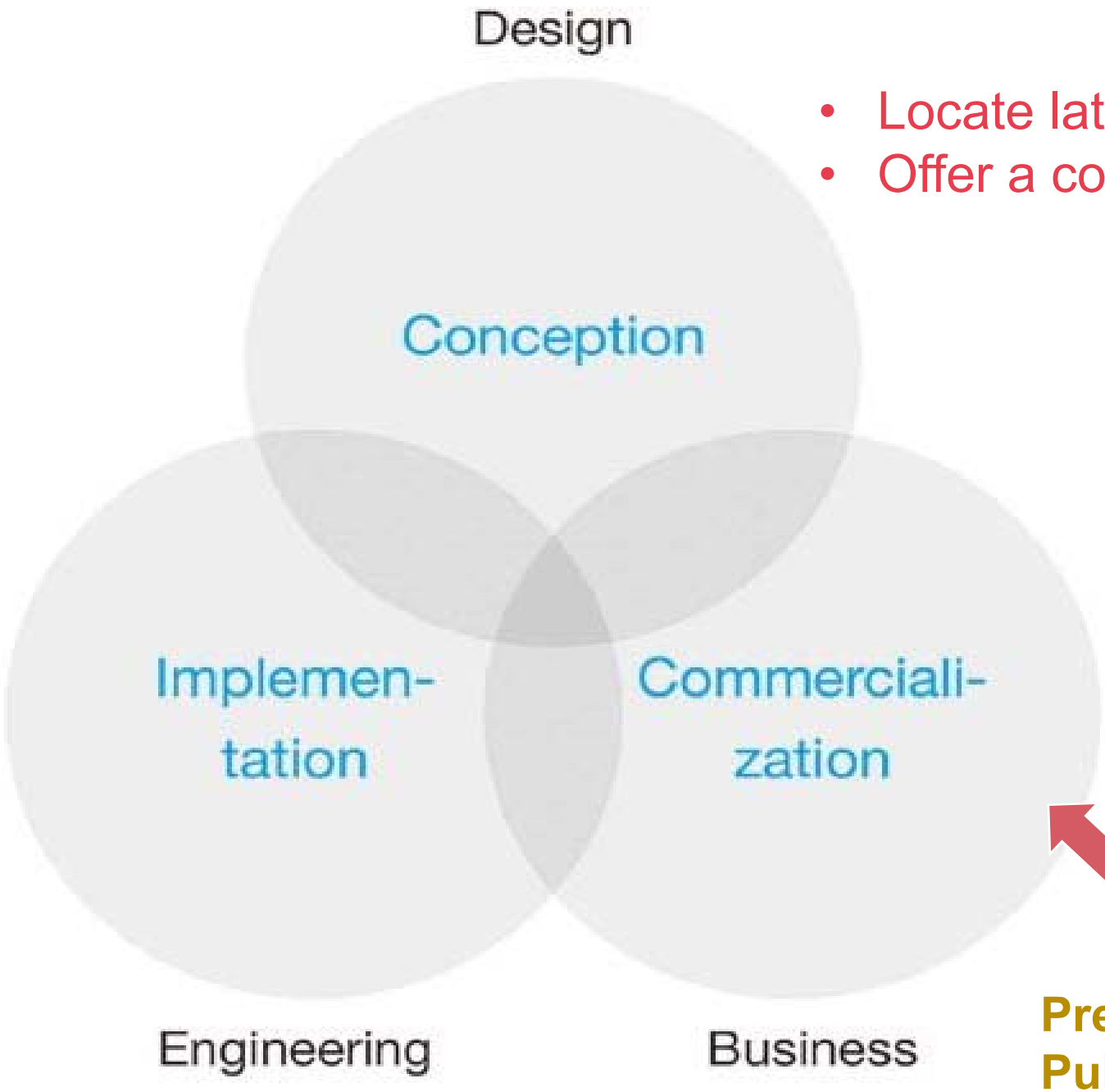
On a 5-point scale, where 5 equals extremely important," here are the top ten skills and qualities, according to employers, and the scores they earned:

SKILL OR ABILITY	SCORE
Ability to verbally communicate with persons inside and outside the organization	4.63
Ability to work in a team structure	4.60
Ability to make decisions and solve problems	4.51
Ability to plan, organize, and prioritize work	4.46
Ability to obtain and process information	4.43
Ability to analyze quantitative data	4.30
Technical knowledge related to the job	3.99
Proficiency with computer software programs	3.95
Ability to create and/or edit written reports	3.56
Ability to sell or influence others	3.55

The book ***Engineering Speaking***

concentrates only on “formal” engineering presentations:

- in-class presentations
- thesis or dissertation defenses
- presentations at professional conferences
- presentations to potential customers or funding agencies.



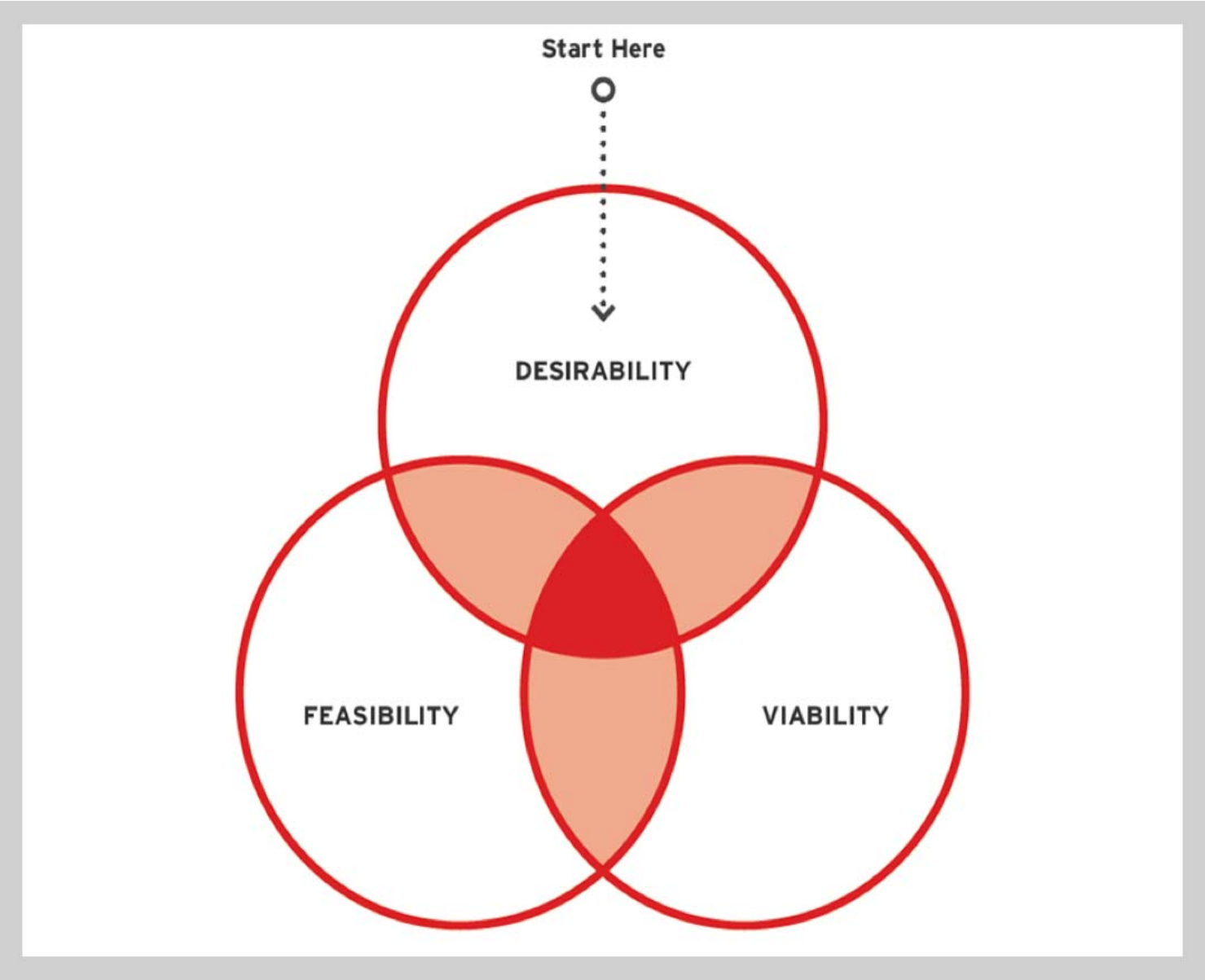
- Locate latent needs
- Offer a concrete solution

Presentation
Public speaking

“Hybrid” professionals

- Highly proficient writers, analytical, and tech savvy, with a strong grasp on business, IT and human behavior.
- “Next-generation” IT/OT Professionals

Hybrid professionals



Example: Too Much Text!

Hertzian Dipole Antenna Co.

These are the attractive properties of leaky-wave antennas

- Leaky wave antennas are much cheaper than phased array antennas. Phased array antennas require separate phase shifters and attenuators which must often be integrated into the antenna aperture – this is expensive!
- Leaky-wave antenna may be constructed on simple planar surfaces like circuit boards. This avoids needing cumbersome 3-dimensional structures which can cause a lot of problems.
- Leaky-wave antennas may be bent or shaped to be conformal to the surface of many types of vehicles, such as aircraft or ground vehicles. This helps improve the aerodynamic performance of the vehicles, and reduces the potential for damaging the antenna. The electrical properties of the antenna might be changed, though.
- Leak-wave antennas can be made with a higher gain than lots of types of antennas such as patches. The gain depends on the length of the leaky-wave antenna in wavelengths.
- Leaky-wave antennas are broadband due to the propagating wave nature of the current on the antenna. They may also be scanned in the elevation plane by changing the frequency within the broad band.
- Leaky-wave antennas are easy to fabricate using simple circuit-board techniques. They may be etched right onto the surface of the board and connectors may be soldered directly to the board.

“Slideuments”

Treating slide sets as stand-alone documents that can be read on their own, without a presenter.

Many researchers say:

“My slides are not there to be beautiful,”

“they are there so that I can share my research and get feedback on my data and methods.”

Presenting ≠ writing

Example: Nice Amount of Text

Hertzian Dipole Antenna Co.

Attractive Properties of leaky-wave antennas

- Low cost alternative to phased arrays
- Planar
- Conformal
- High gain
- Broadband (frequency scanning)
- Easy to fabricate

Presenting ≠ writing

Presenting ≠ writing

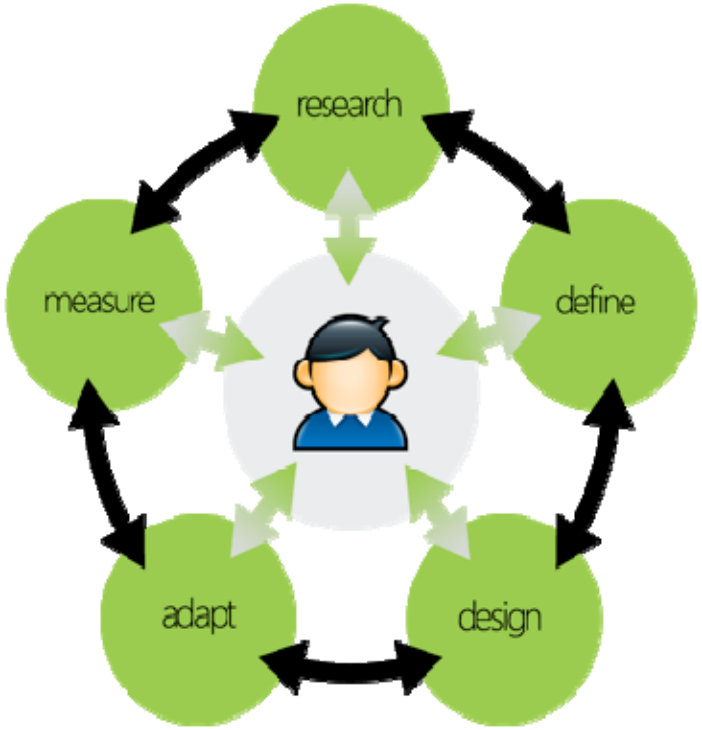
- Audience-centric presentation
  - Understand-ability
  - Share-ability

Unique presentation challenges:

- Presenting highly technical findings and data

Me Me Me VS. You-focus

"*I'm here to save you from your helplessness and ignorance. If you only knew what I know, the world would be better place*."



Audience-centric approach

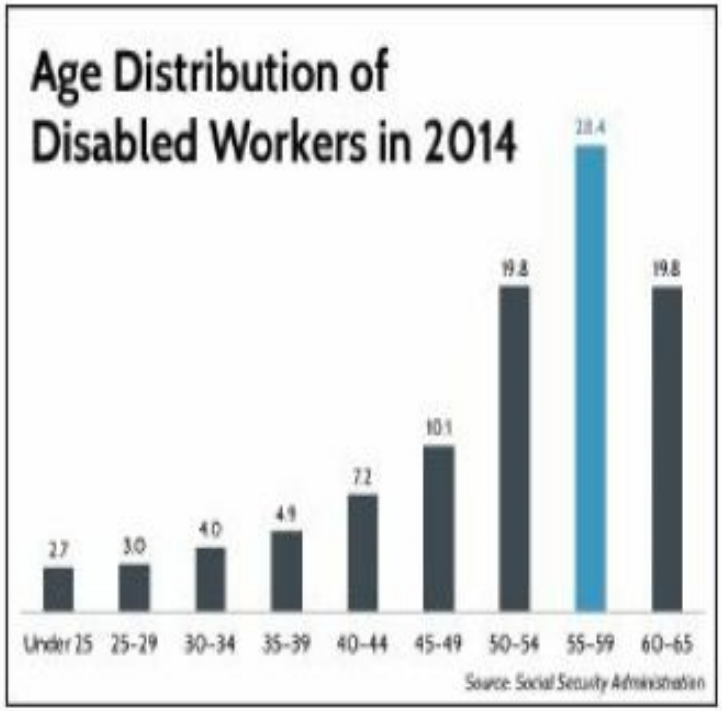
3 driving principles:

1. Visualize your content
2. Unify objects on your slides and throughout your presentation
3. Focus your audience's attention where you want it

1. Visualize your content

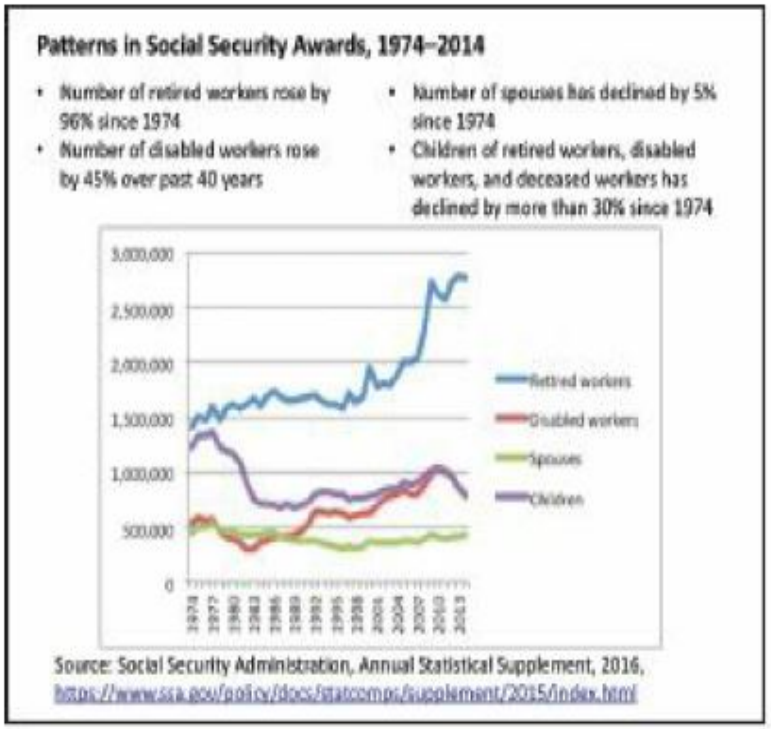
Number of Disabled Workers in 2014

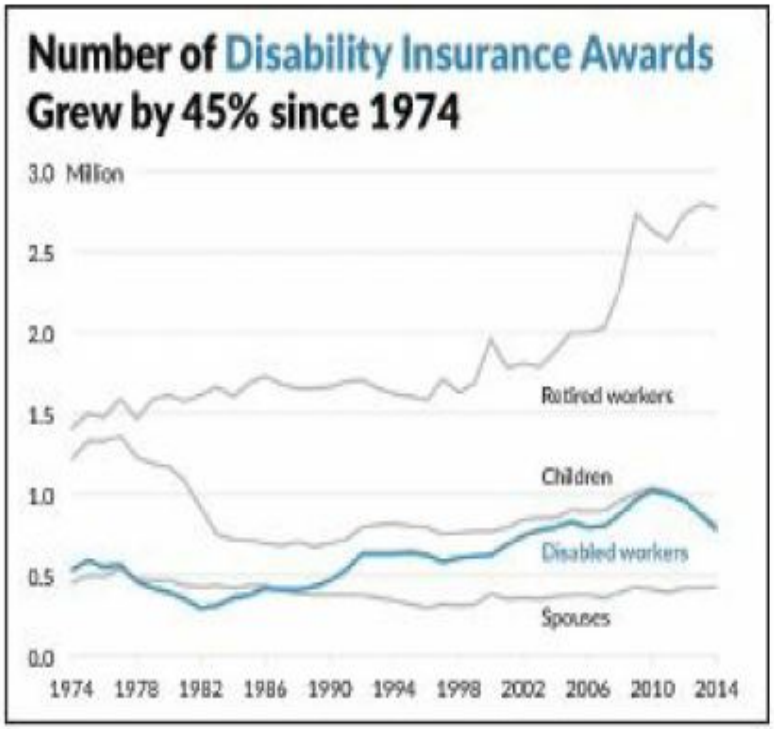
- There was less than 114,000 disabled workers under the age of 40 in 2014
- About 20% of total disabled workers were between 50 and 54 years old
- Another 28% were between 55 and 59 years old
- About 20% were between 60 and 65
  - Some of these workers moved from the disability program to the retirement program



2. Unify objects on your slides and throughout your presentation

3. Focus your audience's attention where you want it



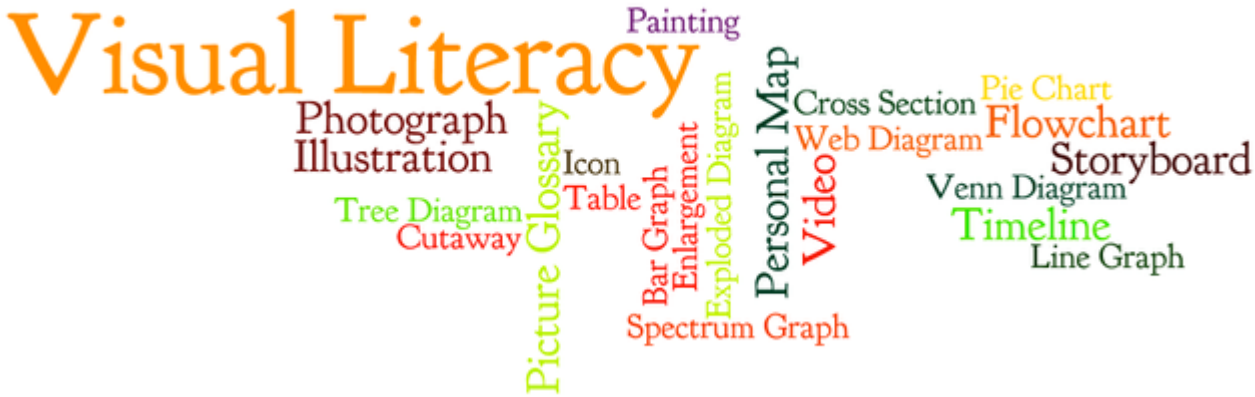


Increasingly popular:

Presentation methods that involve drastically decreasing the amount of words and using abstract illustrations in their place

Visual literacy = Ability to create & interpret visual messages

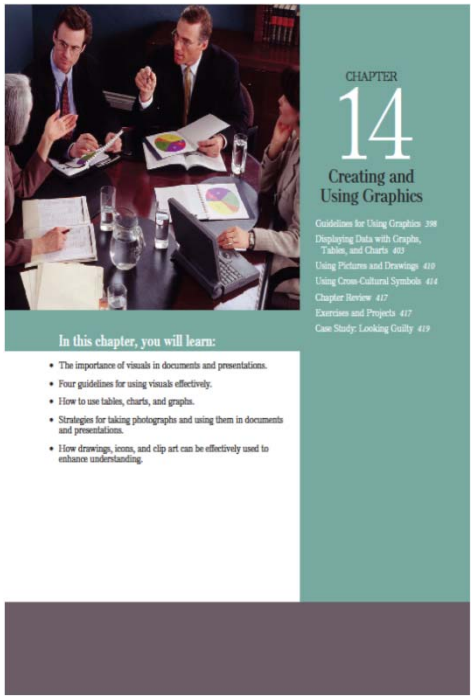
Data
Information
Concepts
Ideas



Visual communication

- Communication design
- Information design
- Graphic design

Reference: Creating and using graphics



CHAPTER

14

Creating and Using Graphics

Guidelines for Using Graphics 398
Displaying Data with Graphs, Tables, and Charts 403
Using Pictures and Drawings 410
Using Cross-Cultural Symbols 414
Chapter Review 417
Exercises and Projects 417
Case Study: Looking Guilty 419

In this chapter, you will learn:

- The importance of visuals in documents and presentations.
- Four guidelines for using visuals effectively.
- How to use tables, charts, and graphs.
- Strategies for taking photographs and using them in documents and presentations.
- How drawings, icons, and clip art can be effectively used to enhance understanding.

The Best Data Visualization Tools of 2017

https://www.pcmag.com/roundup/346417/the-best-data-visualization-tools

The 38 best tools for data visualization

http://www.creativebloq.com/design-tools/data-visualization-712402

A periodic table of visualization methods

http://www.visual-literacy.org/periodic_table/periodic_table.html

10 Free Data Visualization Tools

http://www.pcmag.com/article/347315/10-free-data-visualization-tools

Matching graphics to objectives

Communication Needs → Effective visual choices

Content
Visual impact
Delivery

Thank you!

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